

Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Sunday 21 June 2026

1.1 Daily Limit

Limit of an Infinite Series of Integrals

Evaluate the limit:

$$\lim_{n \rightarrow \infty} \left(\sum_{r=0}^n \left(\frac{1}{2r+1} \int_0^1 \sin((2r+1)x) dx \right) \right)$$

Solution: First, we evaluate the definite integral with respect to x :

$$\int_0^1 \sin((2r+1)x) dx = \left[-\frac{\cos((2r+1)x)}{2r+1} \right]_0^1 = \frac{1 - \cos(2r+1)}{2r+1}$$

Substitute this result back into the summation. Since we are taking the limit as $n \rightarrow \infty$, we rewrite the expression as an infinite series:

$$L = \sum_{r=0}^{\infty} \frac{1 - \cos(2r+1)}{(2r+1)^2}$$

We can split this into two separate series:

$$L = \sum_{r=0}^{\infty} \frac{1}{(2r+1)^2} - \sum_{r=0}^{\infty} \frac{\cos(2r+1)}{(2r+1)^2}$$

The first series is a known result derived from the Basel problem. The sum of the reciprocals of odd squares evaluates to:

$$\sum_{r=0}^{\infty} \frac{1}{(2r+1)^2} = \frac{\pi^2}{8}$$

To evaluate the second series, we consider the Fourier series expansion of the absolute value function $f(x) = |x|$ on the interval $[-\pi, \pi]$:

$$|x| = \frac{\pi}{2} - \frac{4}{\pi} \sum_{r=0}^{\infty} \frac{\cos((2r+1)x)}{(2r+1)^2}$$

Evaluating this Fourier series at $x = 1$ (which safely lies in the interval $[-\pi, \pi]$), we obtain:

$$1 = \frac{\pi}{2} - \frac{4}{\pi} \sum_{r=0}^{\infty} \frac{\cos(2r+1)}{(2r+1)^2}$$

Rearranging the equation to solve for our second series gives:

$$\sum_{r=0}^{\infty} \frac{\cos(2r+1)}{(2r+1)^2} = \frac{\pi}{4} \left(\frac{\pi}{2} - 1 \right) = \frac{\pi^2}{8} - \frac{\pi}{4}$$

Finally, substitute both evaluated series back into the equation for L :

$$L = \frac{\pi^2}{8} - \left(\frac{\pi^2}{8} - \frac{\pi}{4} \right) = \frac{\pi}{4}$$

Monday 22 June 2026

2.1 Daily Limit

Limit of a Sequence of Functions

Evaluate the limit:

$$\lim_{n \rightarrow \infty} \int_0^1 \frac{1}{1 + (x \sec(x))^n} dx$$

Solution: Let $f_n(x) = \frac{1}{1 + (x \sec(x))^n}$. To find the limit of this integral, we examine the behavior of the base function $g(x) = x \sec(x) = \frac{x}{\cos(x)}$ on the interval $[0, 1]$. We first analyze the monotonicity of $g(x)$. Taking the derivative yields:

$$g'(x) = \sec(x) + x \sec(x) \tan(x)$$

For $x \in [0, 1]$, both $\sec(x)$ and $\tan(x)$ are strictly positive (except at 0 where the derivative is 1), hence $g'(x) > 0$. This implies $g(x)$ is strictly increasing on $[0, 1]$.

Since $g(0) = 0$ and $g(1) = \sec(1) \approx 1.85 > 1$, by the Intermediate Value Theorem there exists a unique value $\alpha \in (0, 1)$ such that $g(\alpha) = 1$. This implies $\alpha \sec(\alpha) = 1 \implies \cos(\alpha) = \alpha$. The constant α is known as the Dottie number ($\alpha \approx 0.739$).

Now we analyze the pointwise limit of $f_n(x)$ as $n \rightarrow \infty$:

- For $0 \leq x < \alpha$, we have $0 \leq g(x) < 1$. Thus, $(g(x))^n \rightarrow 0$ as $n \rightarrow \infty$, which gives $\lim_{n \rightarrow \infty} f_n(x) = \frac{1}{1+0} = 1$.
- For $\alpha < x \leq 1$, we have $g(x) > 1$. Thus, $(g(x))^n \rightarrow \infty$ as $n \rightarrow \infty$, which gives $\lim_{n \rightarrow \infty} f_n(x) = 0$.

Since $|f_n(x)| \leq 1$ for all $x \in [0, 1]$ and all n , we can invoke the Dominated Convergence Theorem to interchange the limit and the integral:

$$\lim_{n \rightarrow \infty} \int_0^1 f_n(x) dx = \int_0^1 \left(\lim_{n \rightarrow \infty} f_n(x) \right) dx$$

Splitting the integral at $x = \alpha$:

$$= \int_0^\alpha 1 dx + \int_\alpha^1 0 dx = [x]_0^\alpha + 0 = \alpha$$

Thus, the limit evaluates to α , where α is the unique real root of $x = \cos(x)$.

Tuesday 23 June 2026

3.1 Daily Limit

Limit of an Integral with Rational Functions

Evaluate the limit:

$$\lim_{n \rightarrow \infty} \left(\int_0^\infty \frac{\sqrt{n}}{(x^2 + 1)^n} dx \right)$$

Solution: To evaluate this integral, we apply a substitution that scales with n . Let $u = x\sqrt{n}$. The differential is $du = \sqrt{n} dx$, which means $dx = \frac{du}{\sqrt{n}}$. Substituting these into the integral gives:

$$\int_0^\infty \frac{\sqrt{n}}{\left(\left(\frac{u}{\sqrt{n}}\right)^2 + 1\right)^n} \frac{du}{\sqrt{n}} = \int_0^\infty \frac{1}{\left(1 + \frac{u^2}{n}\right)^n} du$$

Now we pass the limit as $n \rightarrow \infty$ inside the integral. We justify this via the Dominated Convergence Theorem. For $n \geq 1$, by Bernoulli's inequality, $(1 + u^2/n)^n \geq 1 + n(u^2/n) = 1 + u^2$. Thus, the sequence of functions $f_n(u) = (1 + u^2/n)^{-n}$ is bounded above by the dominating function $\frac{1}{1+u^2}$, which is integrable on $[0, \infty)$ since $\int_0^\infty \frac{1}{1+u^2} du = \frac{\pi}{2}$.

Recall the limit definition of the exponential function:

$$\lim_{n \rightarrow \infty} \left(1 + \frac{z}{n}\right)^n = e^z$$

Applying this to our integrand with $z = -u^2$, we have:

$$\lim_{n \rightarrow \infty} \frac{1}{\left(1 + \frac{u^2}{n}\right)^n} = e^{-u^2}$$

By swapping the limit and the integral, we get a standard Gaussian integral over half the real line:

$$\int_0^\infty e^{-u^2} du$$

It is well-known that the full Gaussian integral evaluates to $\int_{-\infty}^\infty e^{-u^2} du = \sqrt{\pi}$. Since the integrand e^{-u^2} is an even function, the integral over the positive real axis is exactly half of the full integral. Thus, the limit evaluates to:

$$\frac{\sqrt{\pi}}{2}$$

Wednesday 24 June 2026

4.1 Daily Limit

Double Limit of a Double Sum

Evaluate the limit:

$$\lim_{m \rightarrow \infty} \lim_{n \rightarrow \infty} \sum_{k=1}^m \sum_{r=kn+1}^{(k+1)n} \frac{n}{r^2}$$

Solution: We begin by evaluating the inner limit with respect to n . Consider the inner sum:

$$S_n(k) = \sum_{r=kn+1}^{(k+1)n} \frac{n}{r^2} = \frac{1}{n} \sum_{r=kn+1}^{(k+1)n} \frac{1}{(r/n)^2}$$

As $n \rightarrow \infty$, this sum forms exactly a Riemann sum. The interval of summation spans from $r \approx kn$ to $r \approx (k+1)n$, which corresponds to the continuous bounds $x = k$ to $x = k+1$ under the partition spacing $\Delta x = \frac{1}{n}$ where $x_i = \frac{r}{n}$. Therefore, the inner limit converges to the definite integral:

$$\lim_{n \rightarrow \infty} S_n(k) = \int_k^{k+1} \frac{1}{x^2} dx$$

Evaluating this definite integral, we get:

$$\int_k^{k+1} x^{-2} dx = \left[-\frac{1}{x} \right]_k^{k+1} = -\frac{1}{k+1} - \left(-\frac{1}{k} \right) = \frac{1}{k} - \frac{1}{k+1}$$

Now, substitute this result back into the outer limit:

$$\lim_{m \rightarrow \infty} \sum_{k=1}^m \left(\frac{1}{k} - \frac{1}{k+1} \right)$$

This sum is a telescoping series. Writing out the first few terms makes the cancellation apparent:

$$\left(1 - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \cdots + \left(\frac{1}{m} - \frac{1}{m+1} \right)$$

All intermediate terms cancel perfectly, leaving only the first and last terms:

$$\sum_{k=1}^m \left(\frac{1}{k} - \frac{1}{k+1} \right) = 1 - \frac{1}{m+1}$$

Finally, taking the limit as $m \rightarrow \infty$:

$$\lim_{m \rightarrow \infty} \left(1 - \frac{1}{m+1} \right) = 1 - 0 = 1$$

Thursday 25 June 2026

5.1 Daily Limit

Limit of an Arc Length

Find the limit of the total length of the curve $y = x^n$ on the interval $[-1, 1]$ as $n \rightarrow \infty$.

Solution: The formula for the arc length of a curve $y = f(x)$ on an interval $[a, b]$ is given by:

$$L_n = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

For $f(x) = x^n$, the derivative is $\frac{dy}{dx} = nx^{n-1}$. Therefore, the length on $[-1, 1]$ is:

$$L_n = \int_{-1}^1 \sqrt{1 + (nx^{n-1})^2} dx$$

Notice that the integrand is an even function regardless of whether n is even or odd, due to the square on the derivative. Thus, we can exploit symmetry:

$$L_n = 2 \int_0^1 \sqrt{1 + n^2 x^{2n-2}} dx$$

To evaluate the limit as $n \rightarrow \infty$, we analyze the sequence geometrically. The curve $y = x^n$ for $x \in [0, 1]$ exhibits specific asymptotic behavior. For any point $x \in [0, 1)$, $x^n \rightarrow 0$ as $n \rightarrow \infty$. At exactly $x = 1$, $x^n = 1$. Geometrically, the curve is "flattening" against the x-axis for almost the entire interval, then making an increasingly sharp turn near $x = 1$ to reach the point $(1, 1)$. In the limit, the path converges to two straight line segments:

1. A horizontal segment along the x-axis from $(0, 0)$ to $(1, 0)$.
2. A vertical segment from $(1, 0)$ to $(1, 1)$.

The length of the first segment is 1, and the length of the second segment is 1. The length of the curve on $[0, 1]$ approaches the sum of these segments: $1 + 1 = 2$. To prove this analytically, we bound the arc length integral: Since $\sqrt{1 + A^2} \leq 1 + A$ for $A \geq 0$, we establish an upper bound:

$$\int_0^1 \sqrt{1 + (nx^{n-1})^2} dx \leq \int_0^1 (1 + nx^{n-1}) dx = [x + x^n]_0^1 = 2$$

For a lower bound, let $\epsilon \in (0, 1)$. We split the integral:

$$\int_0^{1-\epsilon} \sqrt{1 + (nx^{n-1})^2} dx + \int_{1-\epsilon}^1 \sqrt{1 + (nx^{n-1})^2} dx$$

Using $\sqrt{1 + A^2} \geq 1$ for the first part, and $\sqrt{1 + A^2} \geq A$ for the second part:

$$\geq \int_0^{1-\epsilon} 1 dx + \int_{1-\epsilon}^1 nx^{n-1} dx = (1 - \epsilon) + [x^n]_{1-\epsilon}^1 = (1 - \epsilon) + (1 - (1 - \epsilon)^n)$$

As $n \rightarrow \infty$, $(1 - \epsilon)^n \rightarrow 0$, so the lower bound approaches $2 - \epsilon$. Since this holds for arbitrarily small $\epsilon > 0$, the limit of the integral on $[0, 1]$ must be exactly 2. Since the curve spans $[-1, 1]$, by symmetry, the total length evaluates exactly to:

$$2 \times 2 = 4$$